

Optimal Event Trading

Introduction

Event driven hedge funds rely on forming views on catalysts to generate PnL. In the case of earnings, a PM can generate 25% to 50% of their PnL from these bets. Thus, it is important to develop a framework for positioning and trading around events.

Market Impact Model

To specify the cost of trading, we adopt the market-impact framework of Almgren and Chriss [1]. Temporary price impact is taken to be linear in the trading rate: executing at rate $\dot{x}(t)$ displaces the execution price by an amount proportional to $\dot{x}(t)$. Because the cost accrued over an instant is the product of this displacement and the amount traded, it is quadratic in the rate.

At calendar time t you trade at a dollar rate $\dot{x}(t)$ against a market dollar volume rate $v(t)$; the participation rate is

$$\pi(t) = \dot{x} \frac{t}{v(t)}.$$

Heuristic Model

We begin by developing a minimal model that isolates the essential trade-off governing event-driven position sizing. The trade is reduced to two competing terms: the expected profit accruing from the position held on the event date, $\alpha(t)x(t)$, and the transaction cost incurred in establishing that position. Maximizing their difference yields a closed-form expression for the optimal size. Despite its simplicity, the model recovers the standard heuristics used by practitioners and provides quantitative guidance that we refine in the more general treatment that follows.

References

- [1] R. Almgren and N. Chriss, “Optimal execution of portfolio transactions,” *Journal of Risk*, vol. 3, no. 2, pp. 5–39, 2000.

Appendix

Notation

symbol	meaning	units
$v(t)$	market's instantaneous dollar volume rate — the rate at which the tape prints, at calendar time t	\$/day
$\tau(t) = \int_0^t v(s) \, ds$	cumulative market dollars traded since $t = 0$	\$
V	a fixed reference daily volume — ADV, used only to rescale	\$/day
$\dot{x}(t)$	your dollar trading rate	\$/day
$x(t)$	position, in dollars	\$
$\alpha(t)$	expected return over the event	dimensionless
$\pi(t) = \dot{x}(t)/v(t)$	participation rate — your trading rate as a fraction of market volume	dimensionless